

Assignment Code: DS-AG-005

Statistics Basics| Assignment

Instructions: Carefully read each question. Use Google Docs, Microsoft Word, or a similar tool to create a document where you type out each question along with its answer. Save the document as a PDF, and then upload it to the LMS. Please do not zip or archive the files before uploading them. Each question carries 20 marks.

Total Marks: 200

Question 1: What is the difference between descriptive statistics and inferential statistics? Explain with examples.

Answer:

1. Descriptive Statistics

These are used to describe or summarize the characteristics of a specific dataset. They don't try to generalize beyond the data being analyzed.

- Goal: To provide a clear overview of the data through numbers or graphs.
- Common Measures: Mean (average), Median, Mode, Range, and Standard Deviation.
- Example: A teacher calculates the average score of their class on a math test. If the average is 85%, that "85%" is a descriptive statistic describing *only* that specific class.

2. Inferential Statistics

These are used to take a small sample of data and make inferences (predictions or generalizations) about the larger population from which the sample was drawn.

- Goal: To draw conclusions that extend beyond the immediate data.
- Common Measures: Hypothesis testing (t-tests, ANOVA), P-values, and Confidence Intervals.
- Example: A political pollster surveys 1,000 voters to predict who will win a national election. They use the data from those 1,000 people to "infer" how the entire country (millions of people) will likely vote.

Question 2: What is sampling in statistics? Explain the differences between random and stratified sampling.

Answer:

Sampling in Statistics

Sampling is the process of selecting a smaller group (sample) from a larger population to collect data and draw conclusions about the entire population. It saves time, cost, and effort compared to studying every member of the population.

Difference Between Random Sampling and Stratified Sampling

Random Sampling	Stratified Sampling
Every member of the population has an equal chance of being selected.	The population is divided into groups (strata) based on common characteristics, and samples are taken from each group.
Selection is completely random.	Selection is random within each stratum.
Suitable when the population is relatively homogeneous.	Suitable when the population contains different subgroups.
Simpler to implement.	More complex but often provides more accurate results.

Example

Random Sampling:

A college has 1,000 students. A researcher randomly selects 100 students from the entire college.

Stratified Sampling:

The same college has 600 male and 400 female students. The researcher divides students into male and female groups and then randomly selects students from each group in proportion to their size.

Question 3: Define mean, median, and mode. Explain why these measures of central tendency are important.

Answer:

The image asks for definitions of the mean, median, and mode, as well as an explanation of their importance.

Definitions

- **Mean:** Often called the average, it is calculated by adding all the values in a data set and dividing the sum by the total number of values.
- **Median:** The middle value in a data set when the numbers are arranged in order from least to greatest. If there is an even number of values, it is the average of the two middle numbers.
- **Mode:** The value that appears most frequently in a data set. A set can have one mode, more than one mode (multimodal), or no mode at all.

Importance of Measures of Central Tendency

Measures of central tendency are essential because they provide a single value that represents the center or "typical" value of a data set. Their importance includes:

- **Data Summarization:** They allow you to summarize large amounts of data into one understandable figure.
- **Comparison:** They make it easier to compare different data sets (e.g., comparing the average test scores of two different classes).
- **Identifying Skewness:** By comparing the mean, median, and mode, researchers can determine if data is symmetrical or skewed (pulled toward one end by outliers).
- **Decision Making:** They help in practical scenarios, such as a business identifying its most popular product (mode) or a government determining the typical household income (median) to set policy.

Question 4: Explain skewness and kurtosis. What does a positive skew imply about the data?

Answer:

Skewness

Skewness measures the asymmetry of a probability distribution. It indicates whether the data is concentrated on one side of the mean.

- **Symmetric:** Data is evenly distributed (like a Normal distribution).
- **Negative Skew (Left-skewed):** The left "tail" is longer; most data points are clustered on the right.

Kurtosis

Kurtosis measures the "tailedness" of a distribution. It describes how much of the data resides in the tails relative to the peak.

- **High Kurtosis:** Heavy tails and a sharp peak (indicates frequent outliers).
- **Low Kurtosis:** Light tails and a flat peak (indicates fewer outliers).

What does a positive skew imply?

A positive skew (also known as right-skewed) implies the following about the data:

- **Long Right Tail:** The distribution has a long tail stretching toward the higher values on the right.
- **Concentration:** Most of the data points are clustered on the left (lower end) of the scale.
- **Relationship of Measures:** The mean is typically greater than the median because it is pulled toward the extreme values in the right tail:

$$\text{Mean} > \text{Median} > \text{Mode}$$

Question 5: Implement a Python program to compute the mean, median, and mode of a given list of numbers.

```
numbers = [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28]
```

(Include your Python code and output in the code box below.)

Answer:

Paste your code and output inside the box below:

[Code link](#)

Question 6: Compute the covariance and correlation coefficient between the following two datasets provided as lists in Python:

```
list_x = [10, 20, 30, 40, 50]
```

```
list_y = [15, 25, 35, 45, 60]
```

(Include your Python code and output in the code box below.)

Answer:

Paste your code and output inside the box below:

[Code link](#)

Mathematical Summary

- **Covariance (287.5):** A positive value indicates that as x increases, y also tends to increase.
- **Correlation (0.9939):** A value very close to 1 indicates a very strong positive linear relationship between the two datasets.

Question 7: Write a Python script to draw a boxplot for the following numeric list and identify its outliers. Explain the result:

```
data = [12, 14, 14, 15, 18, 19, 19, 21, 22, 22, 23, 23, 24, 26, 29, 35]
```

(Include your Python code and output in the code box below.)

Answer:

[Code link](#)

Python Script Explanation of Results

1. **Quartiles:**

- The **Median (Q2)** is the middle line in the box (approximately **21.5**).
- The **First Quartile (Q1)** is the left edge of the box (**17.25**).
- The **Third Quartile (Q3)** is the right edge of the box (**23.25**).

2. **Interquartile Range (IQR):** The width of the box is $23.25 - 17.25 = 6.0$.

3. **Outliers:**

- Using the standard rule, any value outside the range $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ is an outlier.
- Lower limit: $17.25 - 9 = 8.25$
- Upper limit: $23.25 + 9 = 32.25$
- **Result:** The value **35** is greater than the upper limit of 32.25. Therefore, **35 is the only outlier** in this dataset.

Question 8: You are working as a data analyst in an e-commerce company. The marketing team wants to know if there is a relationship between advertising spend and daily sales.

- Explain how you would use covariance and correlation to explore this relationship.
- Write Python code to compute the correlation between the two lists:

`advertising_spend = [200, 250, 300, 400, 500]`

`daily_sales = [2200, 2450, 2750, 3200, 4000]`

(Include your Python code and output in the code box below.)

Answer:

Exploring the Relationship

- **Covariance:** Use this to determine the direction of the linear relationship. A positive covariance indicates that as advertising spend increases, daily sales tend to increase. However, because covariance is scale-dependent, its magnitude doesn't tell us how strong the relationship is.
- **Correlation (Pearson's r):** Use this to determine both the direction and strength of the linear relationship. It scales the covariance to a range between -1 and 1. A value close to 1 would suggest a very strong positive relationship,

meaning marketing spend is a highly reliable predictor of sales in this dataset.

[Code link](#)

Question 9: Your team has collected customer satisfaction survey data on a scale of 1-10 and wants to understand its distribution before launching a new product.

- Explain which summary statistics and visualizations (e.g. mean, standard deviation, histogram) you'd use.
- Write Python code to create a histogram using Matplotlib for the survey data:

```
survey_scores = [7, 8, 5, 9, 6, 7, 8, 9, 10, 4, 7, 6, 9, 8, 7]
```

(Include your Python code and output in the code box below.)

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Answer:

1. Summary Statistics and Visualizations

- Summary Statistics:
 - Mean: To find the average satisfaction level.
 - Median: To identify the middle value and check if the data is skewed (less sensitive to outliers than the mean).
 - Mode: To see the most frequent score given by customers.
 - Standard Deviation: To measure the spread or consistency of the scores.
- Visualizations:
 - Histogram: To visualize the frequency of scores and see the shape of the distribution.
 - Box Plot: To identify the range, quartiles, and any potential outliers in the data.

2. Python Code for Histogram

Here is the code using `Matplotlib` to create the histogram for the provided survey

data:

[Code link](#)